

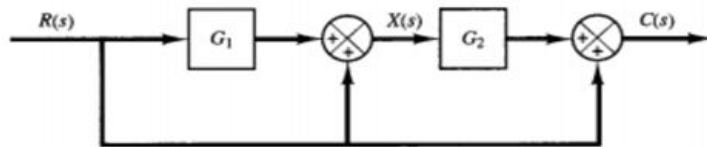
CONTROL SYSTEMS – UE24EE241B

Sample Questions

Block Diagrams

1.

Consider the Block diagram shown below:

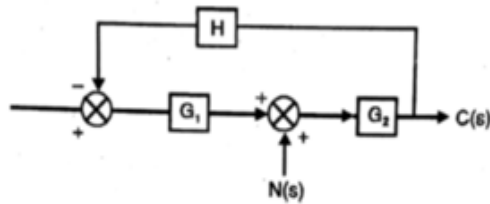


Find the following transfer functions:

1. $\frac{X(s)}{R(s)}$
2. $\frac{C(s)}{R(s)}$

2.

The closed loop system shown in figure is subjected to a disturbance $N(s)$. The transfer function $C(s)/N(s)$ is given by



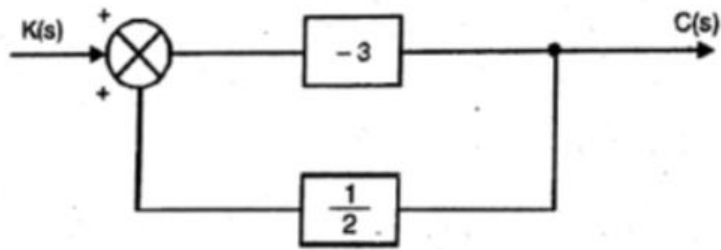
- ☐ $\frac{C(s)}{N(s)} = \frac{G_1}{1 + G_2 G_1 H}$
- ☐ $\frac{C(s)}{N(s)} = \frac{G_2}{1 + G_2 G_1 H}$
- ☐ $\frac{C(s)}{N(s)} = \frac{G_2}{1 + G_2 H}$
- ☐ $\frac{C(s)}{N(s)} = \frac{G_2}{1 + G_2 G_1}$

3.

What is the transfer function if there are 2 forward paths having gains $G_1(s)$ and $G_2(s)$ and no loops?

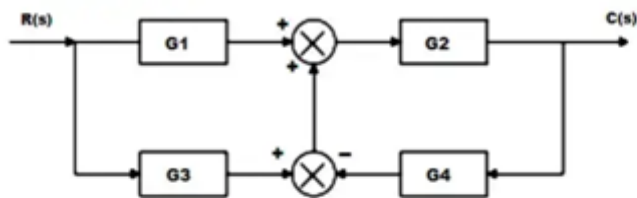
4.

The closed loop gain of the system shown in the given figure is



5.

If the transfer function of the following system is given by $\frac{G_1 G_2 + G_3 G_2}{1 + X}$, what is X ?



Date: 23-01-26

Dr. Anuradha M
(Anchor)